

RAZA SHAUD

Machine Learning | Computer Vision | Embedded Systems | IoT

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SUMMARY

M.Tech student at IIT Guwahati (Systems, Control and Automation) building sensor-driven machine learning and embedded systems end to end. Trained a YOLOv8 cable-fault classifier to 96.5% accuracy with 78% fewer false positives on a solar-powered Arduino robot, and an MFCC voice classifier to 89%. Three published papers including an IEEE Best Paper Award. GATE 2026 All India Rank 2139.

EDUCATION

M.Tech, Electronics and Electrical Engineering — Indian Institute of Technology Guwahati 2026 – 2028
Specialization: Systems, Control and Automation
B.Tech, Electrical Engineering — Ghani Khan Choudhury Institute of Engineering and Technology, Malda 2022 – 2026
CGPA: 7.54 / 10

EXPERIENCE

Machine Learning Intern — Tech Solutions Inc., Kolkata, India May 2025 – Jul 2025
• Optimized 8 machine learning scripts, improving model prediction speed by 30%.
• Tested and validated classification models as part of a 4-person team.
Internship Trainee, Industrial Automation — Steel Authority of India Limited (SAIL), Bokaro, India Nov 2024 – Dec 2024
• Studied plant electrical systems, protection schemes and automation processes across an integrated steel works.
Internship Trainee, Power Generation — Damodar Valley Corporation (DVC), Mejia, India Jun 2024 – Jul 2024
• Documented daily operations, switchgear arrangements and protection settings on an operating thermal generating station.

PROJECTS

Solar-Powered Robotic Cable Monitoring System — B.Tech Major Project Aug 2025 – Apr 2026
• Built an autonomous inspection robot on an Arduino Uno with a 12V/20W solar PV supply and camera-based insulation monitoring over 500 m of cable.
• Trained a YOLOv8 fault classifier on 1,300+ cable images reaching 96.5% accuracy with 78% fewer false positives; tuned the power budget for 8 hours of continuous field operation.
Tools: Arduino Uno, YOLOv8, CNN, OpenCV, Python, Solar PV | [GitHub](#)
Saline Level Monitoring and Alert System (IoT) — Self Project Jan 2025 – Apr 2025
• Interfaced an ultrasonic sensor with an Arduino microcontroller in Embedded C to track fluid level to 2 cm precision; validated over 50 hours with zero missed empty-bottle events.
• Implemented a dual alert path (buzzer and GSM SMS over UART to 3 contacts), cutting simulated response time by 75%.
Tools: Arduino Uno, Embedded C, Ultrasonic Sensor, GSM Module, UART | [GitHub](#)
Human vs. AI Voice Classification System — Self Project Sep 2024 – Nov 2024
• Extracted MFCC features from 1,500+ audio samples with spectral-gating noise reduction (65% less interference) to build a binary classifier at 89% accuracy.
• Deployed a Random Forest on a 15-dimensional feature vector, beating an SVM baseline by 12% under 5-fold cross-validation.
Tools: Python, Scikit-learn, NumPy, MFCC, Digital Signal Processing | [GitHub](#)

TECHNICAL SKILLS

- **Programming:** Python, C, C++
- **Machine Learning & Computer Vision:** YOLOv8, CNN, OpenCV, Scikit-learn, NumPy, Pandas, Random Forest, Feature Extraction
- **Embedded & Hardware:** Arduino Uno, Raspberry Pi, Sensor Interfacing, GSM Module, UART, Hardware Prototyping & Debugging
- **Tools:** MATLAB/Simulink, ANSYS, ETAP, Git, Arduino IDE, Linux, LaTeX, MS Office

PUBLICATIONS

- Dual Rotor Single Stator 6-Phase PMSM for Smart Electric Vehicles. IEEE GIEST 2024 — **Best Paper Award.** [\[Link\]](#)
- Noise Filtration and Voice Discrimination: Case Studies and Results. Tech Research Chronicles (IAST), Vol. 1 Issue 2, 2025. [\[Link\]](#)
- Intelligent Remote Monitoring of Low Voltage Cable Insulation Fault. PEIS-2026, GKCIET Malda, April 2026. [\[Link\]](#)

ACHIEVEMENTS

- GATE 2026 (Electrical Engineering): All India Rank 2139, admitted to M.Tech at IIT Guwahati. 2026
- 100 Days of Code: Complete Python Pro Bootcamp, Udemy Certification. 2024

POSITIONS OF RESPONSIBILITY

- Class Representative, Dept. of Electrical Engineering, GKCIET Malda. 2024 – 2026
- Organizing Committee Member, PEIS-2026 International Conference, GKCIET Malda. Apr 2026